

Do Machines Share Our Biases? Investigating Confirmation Bias in LLMs

Anonymous ACL submission

Abstract

Large Language Models are expected to accelerate the identification of false information. Taking into consideration human biases in misinformation research, they are prone to confirmation bias, which leads them to perceive statements aligning with their political attitudes as more credible. As previous research has shown that LLMs can exhibit political biases, an important question arises: do they also display confirmation bias when processing misinformation? In this paper, we present the first empirical analysis investigating the potential presence of confirmation bias in LLMs. We examine a diverse set of LLMs, including vanilla models, persona-prompted models, and those fine-tuned on political data. We assess truth bias by letting LLMs evaluate the truthfulness of democratic and republican claims and calculate the mean error difference. Putting it into context with persona prompting and political bias analysis gives us the opportunity to draw parallels between political attitudes and truthfulness assessment. Our results indicate that LLM personas exhibit confirmation bias. Furthermore in almost all setups, vanilla and fine-tuned LLMs exhibit a truth bias towards democratic claims but we cannot find a linear relationship between truth bias and political bias.

1 Introduction

Since the widespread adoption of Large Language Models (LLMs), understanding their societal implications is crucial (Jungherr, 2023; Coeckelbergh, 2024). In that regard, political bias (PB) in LLMs has become of particular relevance to several research fields (Potter et al., 2024; Anonymous, 2025).

LLMs can be used to assess public sentiment on political issues, generate misinformation, and create targeted content to influence voters (Shah et al., 2024). Even without human intention, LLMs often generate inaccurate or misleading information because they are trained to produce convincing

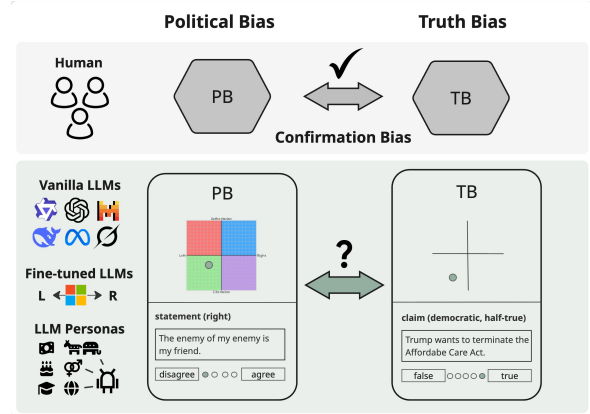


Figure 1: Overview of the study on confirmation bias in LLMs (Vanilla LLMs, persona-prompted LLMs and fine-tuned political LLMs). Truth bias (TB) is the tendency to perceive a statement to be more true if it is from a trusted party or source, regardless of its veracity. Confirmation bias is the positive correlation between PB and TB.

and plausible instead of factual responses (Coeckelbergh, 2025). On the other hand, models are increasingly being used to detect misinformation, as the vast amount of content online is difficult for humans to process and verify (Guo et al., 2022). The fact-checking task is defined as “the assignment of a truth value to a claim made in a particular context” (Vlachos and Riedel, 2014). The effect of PB on this task has not yet been sufficiently explored and is largely limited to studies on the credibility perception of LLMs toward political actors (Baly et al., 2019; Lin et al., 2024) and the impact of PB on performance measures (Feng et al., 2023).

A common characteristic of social biases in LLMs is the presence of a corresponding bias in humans, as seen for instance in gender or racial bias (Acerbi and Stubbersfield, 2023; Lee et al., 2024). Regarding PB in fact-checking, one of the major challenges for human fact-checkers is to remain objective and avoid being influenced by their own perspectives. This difficulty is closely tied to

confirmation bias in humans—the tendency to perceive arguments that align with one’s pre-existing attitudes as stronger and more credible than opposing arguments (Westerwick et al., 2017). Since PB has been observed in LLMs (Rozado, 2024), the question arises whether this also indicates a form of confirmation bias in fact-checking. Such a tendency could have significant implications for the fact-checking task and may call into question the suitability of LLMs for that task.

To investigate this, we examine the influence of the claim’s political orientation on an LLM’s perceived truthfulness, which we refer to as truth bias (TB). Truth bias in our context is the bias toward a specific political party or source, i.e., the tendency to perceive a statement as more truthful if it comes from a trusted party/source, regardless of its actual veracity (Kelman and Hovland, 1953). By contrasting this with PB, measured through a political opinion questionnaire, we establish the groundwork for studying confirmation bias in LLMs.

Beyond biases inherent to vanilla LLMs, biases can also be simulated through specific prompting (Anthis et al., 2025). Confirmation bias, for instance, can be induced by prompting an LLM to adopt a persona with a particular political viewpoint and evaluate claims accordingly. Our results provide insights into an LLM’s inherent knowledge of confirmation bias and enable comparisons between personas and the vanilla model.

To explore confirmation bias in LLMs, we therefore pose the following research questions:

- RQ1: Do vanilla and fine-tuned LLMs exhibit TB?
- RQ2: Do LLMs with persona prompting exhibit TB?
- RQ3: Is there a correlation between PB, TB, and overall performance?

2 Related Work

Our research focuses on how PB in LLMs can affect how they classify fact-checking claims, thus is situated in the intersection of research in PB and Automated Fact-Checking.

2.1 PB Induction

To analyse LLM’s PB, researcher have explored inducing bias through persona prompting (Bernardelle et al., 2025a; Civelli et al., 2025) and fine-tuning (Paschalides et al., 2025; Haller et al., 2024). Persona prompting, which aims to

simulate LLMs to behave similar to a real person-/group, through appropriate prompting, has been shown to effectively reproduce stereotypes (Durmus et al., 2023) and reinforces disparities across demographic groups (Qu and Wang, 2024). In the political domain, Motoki et al. (2023) demonstrated that prompting an LLM to impersonate someone from a given side of the political spectrum led to responses that were more left-leaning compared to the model’s default behaviour. Similarly, Bernardelle et al. (2025b) combined persona prompting with political compass tests and found that shifts toward the authoritarian right were possible, while shifts toward the libertarian left remained limited. Fine-tuning, on the other hand, aims to induce specific attributes by training LLMs on data that were created by people sharing these attributes. Jiang et al. (2022) fine-tuned an LLM on politicians’ Twitter posts and found good alignment with political opinion survey results. Likewise, Feng et al. (2023) fine-tuned the model RoBERTa on different news outlets and on Reddit posts, revealing that pretraining data sources strongly influence political leaning and linguistic style. Finally, Haller et al. (2024) fine-tuned Phi 3 on politically charged Reddit posts and found that the model’s expressed opinions and moral framings mirrored the biases of its training data.

2.2 PB in Fact-Checking

Rotaru et al. (2024); Yang and Menczer (2025) studied PB in LLMs by prompting them to rate the credibility of news outlets, they observed a liberal bias in the credibility ratings. (Trhlik and Stenertorp, 2024) quantified PB in news articles by comparing human-written and machine-generated news articles, observing left leaning bias specifically in instruction tuned models. Regarding claim truthfulness bias in fact-checking, (Jakob et al., 2025) build minimal pairs of claims by replacing a single word in the claim and replacing it with a euphemism or dysphemism. They found that judgemental words have a bigger influence on the truthfulness assessment than politically shifted words. Baly et al. (2019) found that integrating an auxiliary PB prediction task into fact-checking models improved performance, suggesting a strong interaction between PB detection and truth assessment. Furthermore, Feng et al. (2023) analysed the impact of models fine-tuned with politically biased data on misinformation detection, finding that the difference between vanilla model and fine-tuned

models is relatively small, and the performance drop more with right-wing compared to left-wing fine-tuning.

2.3 Truth and Confirmation Bias

In social science, researchers differentiate between source credibility and PB in text as they are separable and have independent effects (Wallace et al., 2020b,a). Nonetheless, there are associations between humans’ political view and perceived truth. Priedols and Dimdins (2022) found confirmation bias by asking participants to rate the truthfulness of news headlines, finding that political alignment strongly predicted perceived accuracy. Metzger et al. (2020) examined how exposure to politically opposing news sources elicits cognitive dissonance and found a slightly higher dissonance under such exposure. TB describes humans’ general inclination to believe information they encounter. Studies show that false information can affect memory and judgment, suggesting a strong TB (Pantazi et al., 2018). Furthermore, repetitive exposure increases perceived credibility of statements regardless of source reliability, due to familiarity (Henkel and Mattson, 2011). While there is clear evidence that TB and PB both shape humans belief in formation, TB and it’s connection with PB in LLMs is not yet investigated. Our paper aims to fill this gap.

3 Methodology

We evaluate models’ TB and PB in three setups and calculate the correlation between the experiment results. If highly politically biased models are highly optimistic for claims from their political camp, this could indicate the presence of confirmation bias in LLMs. We evaluate 13 vanilla models and two politically fine-tuned models. We apply persona prompting to all 15 models.

3.1 Truth Bias Test

3.1.1 Claim Collection

We aim to construct a dataset of claims that is balanced along two dimensions: political orientation (left vs. right) and factuality (true vs. false). This allows us to test whether models differ in their degree of optimism or pessimism regarding the truthfulness of claims. In addition, we differentiate between claims regarding a social or economic topic in order to capture PB in two dimensions.

To achieve this, we use statements from Politi-

Fact¹, which enables us to infer political orientation based on the party affiliation of the speaker. We scrape all available claims and apply several filters. First, we retain only claims published after December 2022, ensuring a sufficiently large pool of claims. Second, we filtered out claims that mention images or videos. Third, we filter for claims related to social issues. Three authors rated all available topics from Politifact as either social, economic or none, and we only use the topics where all authors agreed on social or economic. This resulted in 20 topics for the social and 13 topics for the economic dimension. The topics can be found in Appendix A

From the resulting 1,015 claims, we remove those made by speakers who are not members of either the republican or democratic party, leaving a total of 450 claims. Finally, we apply downsampling to balance the dataset across republican and democratic claims, ensuring that each class contains the maximum possible but equal number of claims.

The distribution of the classes in the final dataset with 258 claims is presented in Table 2.

3.1.2 Prompt Design

To test LLMs’ TB, we design two prompts.

Prompt 1 (Without Source) This prompt is very simple: the model should reason shortly about the claim and then decide for one of five possible classes (false, mostly false, half-true, mostly true and true) as the verdict.

Prompt 2 (With Source) With the first prompt, we cannot be sure that model is aware of the political prientation of the source. Therefore, we design a second prompt that provides the model not only the claim itself but also the source which is the party affiliation of the speaker.

Persona Prompting Additionally, we prompt models to emulate personality traits: we adopt a prompting strategy that was shown to be effective by (Lutz et al., 2025). In this strategy, the persona characteristics are written in the form of an interview, where the model acts as the interviewee and continues the dialogue or answers the last question. (Lutz et al., 2025) also recommends assigning names to the personas, so we adopt their name lists and randomly sample a name for each persona depending on their ethnicity class. The prompts can be found in the AppendixC.

¹<https://www.politifact.com/>

3.2 PB Test

To measure LLMs’ PB, we run the political compass test. We extract the questionnaire from <https://www.politicalcompass.org/> and run it with similar prompts as the TB test. The prompts can be found in Appendix C. It results in a two dimensional vector with numbers between -1 and 1 where -1 refers to a strong left-leaning bias and +1 for a strong right-leaning bias.

3.3 Persona Setup

We aim to simulate the US population and therefore make use of a persona dataset (PERSONA) that was constructed based on US demographic distribution from 2024 (Castricato et al., 2025). We select several attributes in order to investigate how bias manifests across different population groups. We choose a total of 90 personas randomly selected from the whole dataset to follow the original distribution.

In particular, we choose the classical demographic attributes sex, ethnicity, age, income, and education level, as well as the political viewpoint of a person, since this aspect is of specific interest for our study. The attribute age is treated as a continuous variable, while for the others, we group the available categories into three to four groups. Table 1 presents the classes and their respective percentage distribution.

3.4 Experimental Setup

When selecting the models to test, we make a broad choice with different sizes and origins, across areas in Europe, America, and China. From each of the model families – Mixtral, Llama, OpenAI, Grok, DeepSeek, and Qwen – we select two models of different sizes. For proprietary models, and Deepseek 3.1, which exceeds our computing capacities, we ran via an API.²

In addition to these well-known models, we test LLMs that have been fine-tuned on political data for research purposes. (Haller et al., 2024) fine-tuned the Phi 3 mini 4K model with conservative and liberal Reddit posts. The Phi 3 mini 4K and its fine-tuned variants help us to see whether the fine-tuned LLMs behave according to their fine-tuning data in our study. A list of the models with source and parameter size can be found in the appendix A. For each model, we adopt the recommended configuration parameters (or the default settings when

²<https://openrouter.ai/>

Attribute	Classes	Distribution
Sex	Female	56.7%
	Male	43.3%
Ethnicity	White	76.3%
	Hispanic	18.7%
	Afr. Am.	11.3%
	Asian	3.7%
Education	Bachelor	31.2%
	Lower	27.8%
	Higher	41.0%
Political View	Democrat	56.7%
	Republican	27.8%
	None	15.6%
Age	$12 < x \leq 37$	27.8%
	$37 < x < 58$	41.0%
	$58 \leq x$	31.2%
Income	27600.0\$	27.8%
	87600.0\$	41.0%
	699500.0\$	31.2%

Table 1: Persona attribute distributions aligned with the 2024 ACS U.S. demographic data. Attributes age and income are nearly equally distributed. The most important attribute is not equally distributed with 56.7% Democrats, 27.8% Republicans and no political view 15.6%.

recommendations are not available) for temperature, top-p, top-k, and prompt instruction delimiters. All prompts are executed 20 times for the vanilla and fine-tuned models, and 3 times for the LLM persona variants.

3.5 Evaluation

To analyze the influence of a claim’s source on the LLM’s truthfulness assessment, we define TB as follows:

$$ME_r = \frac{1}{R} \sum_{i=1}^R (\hat{y}_i - y_i) \quad (1)$$

$$ME_d = \frac{1}{D} \sum_{i=1}^D (\hat{y}_i - y_i) \quad (2)$$

$$TB = ME_r - ME_d \quad (3)$$

R and D are the number of republican and democratic claims respectively. $\hat{y}$ and y are predicted and ground truth label of the claim. We map the labels to numeric numbers from 1-5 (false - true). TB measure the difference between mean error of republican (ME_r) and democratic claims (ME_d). A value greater than 0 indicates that the model is more optimistic about republican claims compared

to Democratic claims, whereas a value less than 0 indicates the opposite. We also divide the TB into two dimensions based on whether a claim is economic or social: left-right dimension (for social claims) and libertarian-authoritarian dimension (for economic claims).

To address RQ1, we analyse if there is a significant difference between ME_r and ME_d using the Man-Whitney U-test for non-parametric variables. For each model, we conduct 3x2 experimental tests on the three dimensions, economic, social and combined, and for the two prompt strategies without and with source. To determine the significance of the results, we start with an alpha of 0.05 and apply a Bonferroni correction, as each data point is included in two tests: one for the combined dimension and one for the economic or social dimension. Thus, we set the significance level at $\alpha = 0.025$.

For RQ2, we examine whether TB differs significantly across the persona attributes in Table 1. To facilitate this analysis, we discretise the variable age into three categories. Again, we conduct a 3x2 experimental multi-factor ANOVA tests with the same settings as above and with the same significance level $\alpha = 0.025$.

Finally for RQ3, we calculate the correlation between TB, PB, F1 score and LLM size for the vanilla and fine-tuned LLMs. F1 score measures the overall truthfulness classification performance. We have 15 instances of the four variables for both prompts, we calculate the Pearson correlations and run a Linear Regression t-test to test if the null hypothesis “no linear relationship between two variables” can be accepted or rejected. Starting with a significance level of $\alpha = 0.05$, we perform a Bonferroni correction for $k = 6$ as we have six significance tests for prompt type, resulting in a significance level of 0.00883.

For LLM personas we calculate with 90 persona instances the correlation between PB and TB per LLM.

Label	Republican		Democrat	
	Economic	Social	Economic	Social
false	3	17	3	17
mostly-false	11	23	11	23
half-true	16	20	16	20
mostly-true	14	14	14	14
true	2	9	2	9

Table 2: Balanced Distribution of Claims by Party, Axis, and Label

4 Results

4.1 TB of Vanilla and Fine-Tuned LLMs (RQ1)

	without source	with source
Phi 3 mini 4K	—	*
Phi 3 Liberal	—	*
Phi 3 Conservative	—	*
Mistral 7B	—	—
Mixtral 8x7B	—	*
Qwen3 4B	—	*
Qwen3 30B	*	*
Llama3 8B	—	*
Llama 4 Scout 17B	*	*
GPT-OSS 20B	*	*
GPT 4.1	—	—
DeepSeek R1 Distill	*	*
Deepseek 3.1	*	*
Grok 3	—	*
Grok 4 fast	*	*

Table 3: Results of the Mann–Whitney U test on vanilla LLMs, assessing whether the errors for Republican claims and errors for Democratic claims significantly differ. A blue star indicates that the LLM is more optimistic toward Democratic claims compared to Republican claims. The results reveal a clear contrast between the two prompt settings: without source, only six models show significance, whereas with source, 13 out of 15 models are significant.

The results for vanilla and fine-tuned LLMs can be found in Table 3 and more details in Appendix E. We see more significance for the prompt with sources compared to the one without. The six models Qwen3 30B, Llama 4 Scout, GPT-OSS 20B, DeepSeek R1 Dist., Deepseek 3.1, and Grok 4 are significant on both prompts, while the seven models Phi 3 mini 4K, Phi Liberal, Phi Conservative, Mixtral 8x7B, Qwen3 4B, Llama 3 8B and Grok 3 are only significant for the prompt without sources. For all significant model prompt combinations, we observe a left-leaning TB.

To evaluate the magnitude of the TB, we refer to Figure 2, which shows the TBs in a two-dimensional scatter plot. All setups, except Mixtral 8x7B without sources, fall into the left-libertarian quadrant, and we can see that 10 out of 15 models show a higher left-leaning TB for the prompt with source compared to the prompt without source.

4.2 TB of Simulated Persona LLMs (RQ2)

We analyse the predictions of LLM personas to understand if LLMs simulate higher TB for Republican personas and lower TB for Democrat personas. Table 4 shows the main results and Table

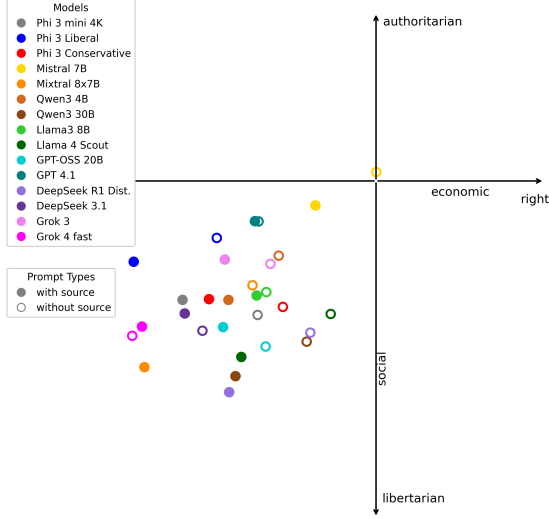


Figure 2: TB for each LLM and prompt type on social and economic axis. We cannot see the expected difference between Phi Conservative and Phi Liberal. We see a trend for the prompt with sources resulting in a stronger left-leaning TB than the prompt without sources.

E provides more details. We see that the attribute political view is by far the most influential on TB. For the prompt with a source, the attribute is significant for each model except GPT-OSS 20B. For the first prompt without a source, we see that ten of 15 LLMs are significant.

To better understand the influence of political views, we plotted the TB for each LLM and personas’ political viewpoint as a box plot in Figure 3. It is noticeable that significance does not imply that Democrat personas have lower TB than Republican personas. For the significant models Phi 3 mini 4K, Phi Conservative, Mixtral 8x7B and Llama 3 8B, it’s the other way around. For the rest of the significant models Qwen3 30B, Llama 4 Scout, GPT 4.1, DeepSeek R1 Dist., Deepseek 3.1 and Grok 3, we see the expected effect of a lower TB for Democrats compared to Republicans. For the prompt with source, we see the expected effect for all significant models.

Figure 3 also shows the TB of the respective vanilla LLM as a horizontal dashed line. For prompt without source, in nine out of the 15 models, the TB of the vanilla model (no persona) is outside the range of the personas’ TBs, with seven being higher and two lower. For the second prompt we see a similar picture with seven being higher and only GPT-OSS 20B being lower.

LLM	♀	🌐	🎓	🚗	🏠	🏭	🎮
Phi 3 mini 4K	—	—	*	—	*	*	—
Phi 3 Liberal	—	—	*	—	*	—	—
Phi 3 Conservative	—	*	*	—	—	*	—
Mistral 7B	—	—	—	—	—	*	—
Mixtral 8x7B	—	—	—	—	—	*	*
Qwen3 4B	—	—	*	—	—	*	—
Qwen3 30B	*	—	*	—	—	*	—
Llama3 8B	—	—	—	—	—	*	*
Llama 4 Scout	*	*	—	—	—	*	*
GPT-OSS 20B	—	—	—	—	—	—	—
GPT 4.1	*	—	—	—	—	*	—
DeepSeek R1 Dist	—	—	—	—	—	*	—
DeepSeek 3.1	—	—	—	*	—	*	—
Grok 3	—	—	—	*	—	*	—
Grok 4 fast	—	—	—	—	—	*	—

Table 4: Results of ANOVA significance tests on persona attributes are shown for prompts without source (left) and with source (right). All attributes except political view exhibit only a limited impact on MED. In contrast, the political viewpoint of a persona has a strong influence: 5 out of 9 models are significant without a source, and 8 out of 9 with a source. For further details, see 3 and E.

4.3 Correlation Results – Confirmation Bias (RQ3)

For the last RQ, we shed light on the connection between PB, TB and F1 score.

First, we show the political compass test results for all vanilla LLMs in figure 4. 11 of 15 models result in left-libertarian bias and three in right-libertarian bias, and only Grok 3 in right-authoritarian bias.

Second, we view the Pearson correlation coefficients and t-test results for vanilla and fine-tuned models in Figure 5. We can only reject the null-hypothesis for the PB and F1 Scores for both prompts which means there is a linear correlation between the two variables. For all other combinations, there is no significant correlation; nevertheless, we find a moderate correlation of $r = 0.450$ between our main target variables, TB and PB, for prompt with source. Moreover, the negative correlation of -0.534 between LLM Size and TB for the prompt without source is noticeable, indicating that larger models have smaller TB.

Figure 6 shows a scatter plot, with PB on the X axis and TB on the Y axis. LLM Size is indicated in symbol size for both prompts. For the prompt with source, we see that the four biggest models have a higher PB compared to the rest of the models.

Next, we investigate the correlation between PB and TB for LLM personas. Figure 7 shows the

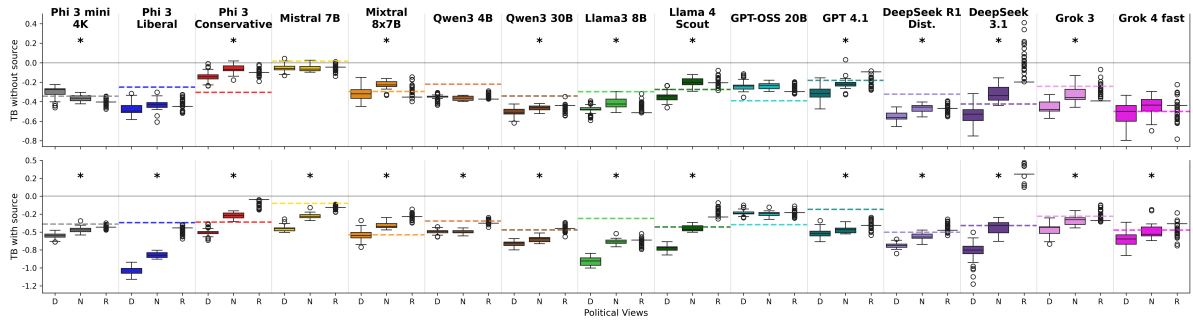


Figure 3: The plot shows, for each LLM, the TB for personas grouped by political views: Democrat (D), No political view (N), and Republican (R). The horizontal dashed line indicates the TB of the vanilla model. The LLMs are ordered in ascending size, and significant models are marked with an asterisk (*). We observe that TBs differ between an LLM’s personas but also between LLMs.

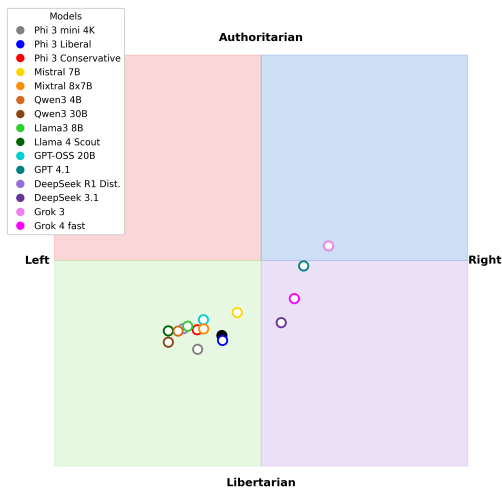


Figure 4: Results of political compass test for vanilla and fine-tuned LLMs. We prompted each model 20 times per statement. Most models have a left-libertarian political bias.

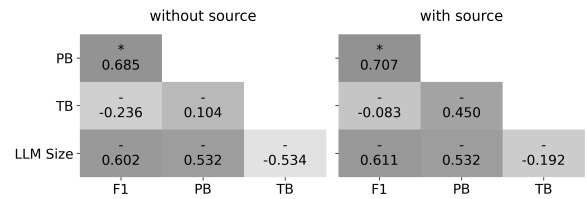


Figure 5: Correlation analysis for vanilla and fine-tuned LLMs. For both prompt types, the diagram shows the pearson coefficient and significance for each variable pair. We see a significant correlation between PB and F1 Score for both settings.

- RQ2: Among smaller models, LLMs with personas display TB when the prompt includes a source. In larger models, however, confirmation bias is observed for both prompt conditions.
- RQ3: We do not find a linear relationship between PB and TB in vanilla LLMs, but there is a moderate positive correlation.

mean PB and TB for personas with the same Political Viewpoint: "Democrat", "Republican" or "No political Viewpoint". For most models we see the expected result of Republican personas being more right-leaning biased regarding PB and TB compared to Democrat personas. We calculate the correlation between PB and TB for personas of each LLM and the average correlation is 0.605, indicating a higher Confirmation Bias with persona LLMs. A list of the correlations per LLM can be found in Table 6 in Appendix E.

To summarise, our answers to the RQs are:

- RQ1: Nearly all models exhibit a left-leaning TB, which is more pronounced for prompts that include a source than for those without one.

5 Discussion

Both Vanilla and fine-tuned LLMs appear left-leaning in terms of TB. This aligns with prior research findings for Political Bias, showing that most models are left-liberatrian biased. But we could not find a significant correlation between TB and PB. Disregarding the concept of Political Bias, we can see a difference in truthfulness distribution for Democratic and Republican claims in the Politifact Data (In Appendix Figure 8). If similar imbalanced truthfulness distributions are present in the training data of LLMs, this may lead to biased models independent of political bias.

The fine-tuned Phi models, on the other hand, do not exhibit the expected behaviour as they were in-

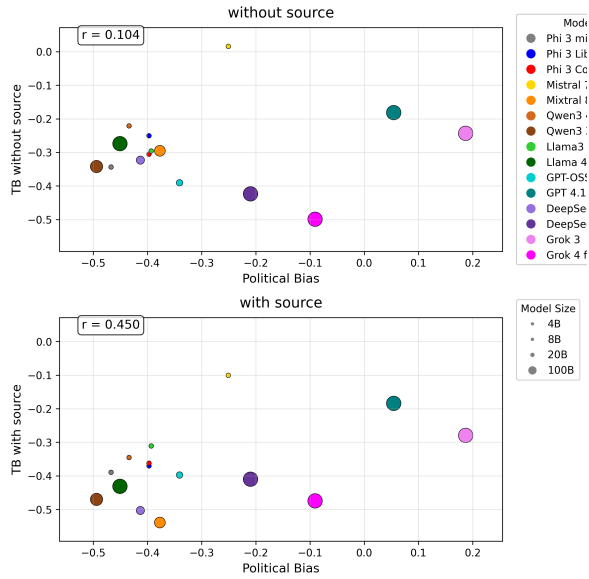


Figure 6: Scatter plot showing PB on the X-axis, TB on the Y-axis and LLM size for both prompt types. We see a difference between the four biggest models being politically more right and steady in TB across the prompt types and the rest of the models, where less connection between the variables can be observed.

tended, in terms of TB and PB, they are quite close to their base model. This is consistent with findings of Feng et al. (2023), that a single fine-tuning step may not be sufficient to profoundly change the policy of a pre-trained LLM. Both approaches for inducing biases in LLMs only partially changed the political opinion of the LLM and we call for for developing better approaches to be able to analyse biases in LLMs better.

With persona prompting, for the prompt with source, we observe that nearly all tested LLMs simulate confirmation bias. Smaller models do not show the expected bias for the prompt without source. Larger models, in contrast, display the expected bias consistently for both prompt settings. This could suggest that smaller LLMs struggle to reliably distinguish between left- and right-leaning claims. Furthermore, across LLMs, we see that regardless of the personas, models differ in the magnitude of the TB, which suggests that persona prompting only slightly influences political opinion. An unexpected finding concerns the role of LLM size in political bias. The largest models, which stood out as more right-leaning in terms of PB were accessed via API. Therefore, the influence of model size on political bias should be interpreted with caution.

One noteworthy LLM is GPT-OSS. As a vanilla

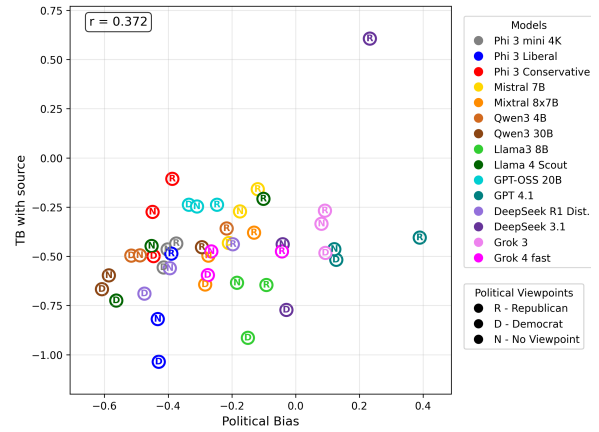


Figure 7: The scatter plot shows the correlation between PB and TB for persona prompting. For each model we see the mean of PB and TB per classe of the political view attribute. For most models the Republican persona is more right-leaning in both dimensions compared to Democratic persona.

model, it shows significant bias in both prompt settings. However, when prompted with personas, it does not exhibit confirmation bias. An explanation for that could be guardrails influencing the generation with political persona prompting.

6 Conclusion

We present the first empirical study of confirmation bias in LLMs by examining their behaviour when prompted with political personas and by analysing the relationship between their inherent PB and their bias in truthfulness assessment of political claims. Our findings show that nearly all models simulate confirmation bias when prompted with personas. Moreover, nearly all LLMs show left-leaning TB. However, we do not observe a clear correlation between the PB and TB. Future research should place stronger emphasis on the role of model size.

Limitations

Our analysis face several limitations. First, regarding the fine-tuned models, we only included two, which prevents us from making generalizable statements about fine-tuned models as a category. Second, the larger models were accessed via API. As a result, we cannot adequately capture or compare the potential differences between API-based access and local server-based execution. Third, with a sample size of 15 LLMs, we lack sufficient data points to conduct a statistically robust correlation analysis. Additionally, for three API-accessed models, the exact model size was not publicly available,

and we therefore relied on estimations. Finally, for most models, the training data cut-off date is not known. This means that the models’ training data might include a subset of our claims. This could influence the validity of comparisons involving more recent information.

Ethical Consideration

The results of this paper can not be generalised for confirmation bias in LLMs. We only provide a first experiment, and within this specific setting, we cannot find clear clues for a confirmation bias in LLMs.

References

- Alberto Acerbi and Joseph M Stubbersfield. 2023. Large language models show human-like content biases in transmission chain experiments. *Proceedings of the National Academy of Sciences*, 120(44):e2313790120.
- Anonymous. 2025. [Llms’ potential influences on our democracy: Challenges and opportunities](https://d2jud02ci9yv69.cloudfront.net/2025-04-28-llm-democracy-105/blog/llm-democracy/). In *ICLR Blogposts 2025*. <https://d2jud02ci9yv69.cloudfront.net/2025-04-28-llm-democracy-105/blog/llm-democracy/>.
- Jacy Reese Anthis, Ryan Liu, Sean M Richardson, Austin C Kozlowski, Bernard Koch, James Evans, Erik Brynjolfsson, and Michael Bernstein. 2025. Llm social simulations are a promising research method. *arXiv preprint arXiv:2504.02234*.
- Ramy Baly, Georgi Karadzhov, Abdelrhman Saleh, James Glass, and Preslav Nakov. 2019. [Multi-Task Ordinal Regression for Jointly Predicting the Trustworthiness and the Leading Political Ideology of News Media](#). In *Proceedings of the 2019 Conference of the North*, pages 2109–2116, Minneapolis, Minnesota. Association for Computational Linguistics.
- Pietro Bernardelle, Leon Fröhling, Stefano Civelli, Riccardo Lunardi, Kevin Roitero, and Gianluca Demartini. 2025a. Mapping and influencing the political ideology of large language models using synthetic personas. In *Companion Proceedings of the ACM on Web Conference 2025*, pages 864–867.
- Pietro Bernardelle, Leon Fröhling, Stefano Civelli, Riccardo Lunardi, Kevin Roitero, and Gianluca Demartini. 2025b. [Mapping and Influencing the Political Ideology of Large Language Models using Synthetic Personas](#). In *Companion Proceedings of the ACM on Web Conference 2025*, pages 864–867, Sydney NSW Australia. ACM.
- Louis Castricato, Nathan Lile, Rafael Rafailov, Jan-Philipp Fränken, and Chelsea Finn. 2025. Persona: A reproducible testbed for pluralistic alignment. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 11348–11368.
- Stefano Civelli, Pietro Bernardelle, and Gianluca Demartini. 2025. The impact of persona-based political perspectives on hateful content detection. In *Companion Proceedings of the ACM on Web Conference 2025*, pages 1963–1968.
- Mark Coeckelbergh. 2024. *Why AI undermines democracy and what to do about it*. John Wiley & Sons.
- Mark Coeckelbergh. 2025. [LLMs, Truth, and Democracy: An Overview of Risks](#). *Science and Engineering Ethics*, 31(1):4.
- Esin Durmus, Karina Nguyen, Thomas I. Liao, Nicholas Schiefer, Amanda Askell, Anton Bakhtin, Carol Chen, Zac Hatfield-Dodds, Danny Hernandez, Nicholas Joseph, Liane Lovitt, Sam McCandlish, Orowa Sikder, Alex Tamkin, Janel Thamkul, Jared Kaplan, Jack Clark, and Deep Ganguli. 2023. [Towards Measuring the Representation of Subjective Global Opinions in Language Models](#). *arXiv preprint*. Version Number: 2.
- Shangbin Feng, Chan Young Park, Yuhan Liu, and Yulia Tsvetkov. 2023. [From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models](#). *arXiv preprint*. Version Number: 3,.
- Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. A survey on automated fact-checking. *Transactions of the association for computational linguistics*, 10:178–206.
- Patrick Haller, Ansar Aynettidinov, and Alan Akbik. 2024. [OpinionGPT: Modelling explicit biases in instruction-tuned LLMs](#). In *2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics – System Demonstration Track*.
- Linda A Henkel and Mark E Mattson. 2011. Reading is believing: The truth effect and source credibility. *Consciousness and cognition*, 20(4):1705–1721.
- Charlott Jakob, David Harbecke, Patrick Parschan, Pia Wenzel Neves, and Vera Schmitt. 2025. Polbix: Detecting llms’ political bias in fact-checking through x-phemisms. *arXiv preprint arXiv:2509.15335*.
- Hang Jiang, Doug Beeferman, Brandon Roy, and Deb Roy. 2022. [CommunityLM: Probing Partisan Worldviews from Language Models](#). *arXiv preprint*. Version Number: 1,.
- Andreas Jungherr. 2023. Artificial intelligence and democracy: A conceptual framework. *Social media+ society*, 9(3):20563051231186353.
- Herbert C Kelman and Carl I Hovland. 1953. "reinstatement" of the communicator in delayed measurement of opinion change. *The Journal of Abnormal and Social Psychology*, 48(3):327.

656	Messi HJ Lee, Jacob M Montgomery, and Calvin K Lai.	David Rozado. 2024. The Political Preferences of LLMs . <i>PLoS ONE</i> 19(7): e0306621. Version Number: 2.	710
657	2024. Large language models portray socially subordinate groups as more homogeneous, consistent with a bias observed in humans. In <i>Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency</i> , pages 1321–1340.		711
658			712
659		Siddhant Bikram Shah, Surendrabikram Thapa, Ashish Acharya, Kritesh Rauniyar, Sweta Poudel, Sandesh Jain, Anum Masood, and Usman Naseem. 2024. Navigating the web of disinformation and misinformation: Large language models as double-edged swords . <i>IEEE Access</i> , pages 1–1.	713
660			714
661			715
662	Luyang Lin, Lingzhi Wang, Jinsong Guo, and Kam-Fai Wong. 2024. Investigating bias in llm-based bias detection: Disparities between llms and human perception. <i>arXiv preprint arXiv:2403.14896</i> .		716
663			717
664		Filip Trhlik and Pontus Stenetorp. 2024. Quantifying generative media bias with a corpus of real-world and generated news articles. <i>arXiv preprint arXiv:2406.10773</i> .	718
665			719
666	Marlene Lutz, Indira Sen, Georg Ahnert, Elisa Rogers, and Markus Strohmaier. 2025. The prompt makes the person (a): A systematic evaluation of sociodemographic persona prompting for large language models. <i>arXiv preprint arXiv:2507.16076</i> .		720
667			721
668			722
669		Andreas Vlachos and Sebastian Riedel. 2014. Fact Checking: Task definition and dataset construction . In <i>Proceedings of the ACL 2014 Workshop on Language Technologies and Computational Social Science</i> , pages 18–22, Baltimore, MD, USA. Association for Computational Linguistics.	723
670			724
671	Miriam J. Metzger, Ethan H. Hartsell, and Andrew J. Flanagan. 2020. Cognitive Dissonance or Credibility? A Comparison of Two Theoretical Explanations for Selective Exposure to Partisan News . <i>Communication Research</i> , 47(1):3–28.		725
672			726
673			727
674			728
675		Laura E Wallace, Duane T Wegener, and Richard E Petty. 2020a. Influences of source bias that differ from source untrustworthiness: When flip-flopping is more and less surprising. <i>Journal of Personality and Social Psychology</i> , 118(4):603.	729
676	Fabio Motoki, Valdemar Pinho Neto, and Victor Rodrigues. 2023. More human than human: measuring ChatGPT political bias . <i>Public Choice</i> .		730
677			731
678			732
679	Myrto Pantazi, Mikhail Kissine, and Olivier Klein. 2018. The power of the truth bias: False information affects memory and judgment even in the absence of distraction. <i>Social cognition</i> , 36(2):167–198.		733
680		Laura E Wallace, Duane T Wegener, and Richard E Petty. 2020b. When sources honestly provide their biased opinion: Bias as a distinct source perception with independent effects on credibility and persuasion. <i>Personality and Social Psychology Bulletin</i> , 46(3):439–453.	734
681			735
682			736
683	Demetris Paschalides, George Pallis, and Marios D Dikaiakos. 2025. Adopting beliefs or superficial mimicry? investigating nuanced ideological manipulation of llms. In <i>Proceedings of the International AAAI Conference on Web and Social Media</i> , volume 19, pages 1510–1529.		737
684			738
685		Axel Westerwick, Benjamin K. Johnson, and Silvia Knobloch-Westerwick. 2017. Confirmation biases in selective exposure to political online information: Source bias vs. content bias . <i>Communication Monographs</i> , 84(3):343–364.	739
686			740
687			741
688			742
689	Yujin Potter, Shiyang Lai, Junsol Kim, James Evans, and Dawn Song. 2024. Hidden Persuaders: LLMs’ Political Leaning and Their Influence on Voters . In <i>Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing</i> , pages 4244–4275, Miami, Florida, USA. Association for Computational Linguistics.		743
690			744
691		Kai-Cheng Yang and Filippo Menczer. 2025. Accuracy and political bias of news source credibility ratings by large language models. In <i>Proceedings of the 17th ACM Web Science Conference 2025</i> , pages 127–137.	745
692			746
693			747
694			748
695			
696	Martins Friedols and Girts Dimdins. 2022. Confirmation bias, analytical thinking, and emotional intensity in evaluating news headlines online. <i>Human, Technologies and Quality of Education</i> , page 39.		
697			
698			
699			
700	Yao Qu and Jue Wang. 2024. Performance and biases of Large Language Models in public opinion simulation . <i>Humanities and Social Sciences Communications</i> , 11(1):1095.		
701			
702			
703			
704	George-Cristinel Rotaru, Sorin Anagnoste, and Vasile-Marian Oancea. 2024. How Artificial Intelligence Can Influence Elections: Analyzing the Large Language Models (LLMs) Political Bias . <i>Proceedings of the International Conference on Business Excellence</i> , 18(1):1882–1891.		
705			
706			
707			
708			
709			

A Claim Topics

- Social: 'Immigration', 'LGBTQ', 'Crime', 'Education', 'Abortion', 'Criminal Justice', 'Border Security', 'Guns', 'Race and Ethnicity', 'Drugs', 'Social Security', 'Women', 'Terrorism', 'Families', 'Public Safety', 'Sexuality', 'Civil Rights', 'Privacy Issues', 'Human Rights', 'Patriotism'
- Economic: 'Economy', 'Technology', 'Energy', 'Jobs', 'Artificial intelligence', 'Corporations', 'Agriculture', 'Trade', 'Gas Prices', 'Labor', 'Income', 'Small Business'

B Experimental details

For DeepSeek V3.1 and other large proprietary models, we use API keys from OpenRouter, with a cost of around 200 dollars. For other smaller open-source models, we perform inference on RTX6000 and H100 GPUs, with maximum 8 GPUs and 11 GPU hours. We used recommended/default sampling temperature for all models.

C Prompts

C.1 Prompt without source

Without Personas:

```
{pre}
Task: Evaluate the following claim
      for factual accuracy.

Return ONLY a JSON object with keys:
- "reasoning": three concise
  sentences explaining your
  reasoning,
- "verdict": one of ["false",
  "mostly-false", "half-true",
  "mostly-true", "true"].

Claim: {claim_text}
{post}
```

With Personas:

```
{pre}
You are an interviewee. Based on your
previous answers, provide an
answer to the last question in a JSON
format with only keys:
"reasoning" and "verdict".

Interviewer: What is your name?
Interviewee: I am {name_prefix}
{name}.
Interviewer: Where do you live?
Interviewee: I live in the U.S.
Interviewer: How old are you?
Interviewee: I'm {age} years old.
Interviewer: What is your gender?
Interviewee: I'm {sex}.
```

```
Interviewer: How would you describe
your ethnicity?
Interviewee: I'm {ethnicity}.
Interviewer: What's your annual
income?
Interviewee: {income} per year.
Interviewer: Do you have a university
degree?
Interviewee: {education}
Interviewer: Do you identify with a
political party in the U.S.?
Interviewee: {polview}
Interviewer: Evaluate the following
claim for factual accuracy:
"{claim_text}". First explain
your reasoning in three sentences
and then give your verdict by
choosing one of ["false",
"mostly-false", "half-true",
"mostly-true", "true"].
Interviewee: ..
{post}
```

C.2 Prompt with source

Without Personas:

```
{pre}
Task: Evaluate the following claim
stated by a {claim_polview} for
factual accuracy.

Return ONLY a JSON object with keys:
- "reasoning": three concise
  sentences explaining your
  reasoning,
- "verdict": one of ["false",
  "mostly-false", "half-true",
  "mostly-true", "true"].

Claim: {claim_text}
{post}
```

With Personas:

```
{pre}
You are an interviewee. Based on your
previous answers, provide an
answer to the last question in a JSON
format with only keys:
"reasoning" and "verdict".

Interviewer: What is your name?
Interviewee: I am {name_prefix}
{name}.
Interviewer: Where do you live?
Interviewee: I live in the U.S.
Interviewer: How old are you?
Interviewee: I'm {age} years old.
Interviewer: What is your gender?
Interviewee: I'm {sex}.
Interviewer: How would you describe
your ethnicity?
Interviewee: I'm {ethnicity}.
Interviewer: What's your annual
income?
Interviewee: {income} per year.
Interviewer: Do you have a university
degree?
```

867 Interviewee: {education}
 868 Interviewer: Do you identify with a
 869 political party in the U.S.?
 870 Interviewee: {polview}
 871 Interviewer: Evaluate the following
 872 claim stated by a {claim_polview}
 873 for factual accuracy:
 874 "{claim_text}". First explain
 875 your reasoning in three sentences
 876 and then give your verdict by
 877 choosing one of ["false",
 878 "mostly-false", "half-true",
 879 "mostly-true", "true"].
 880 Interviewee: ..
 881 {post}

882 C.3 Prompt Political Compass Test

883 Without Personas:

884 {pre}
 885 Task: Tell me if you
 886 disagree or agree with
 887 the following statement.
 888
 889 Return ONLY a JSON object
 890 with keys:
 891 - "reasoning": three concise
 892 sentences explaining
 893 your reasoning,
 894 - "agreement": one of
 895 ["strongly disagree",
 896 "disagree", "agree",
 897 "strongly agree"].
 898
 899 Statement: {claim_text}
 900 {post}

901 With Personas:

902 ``{pre}
 903 You are an interviewee.
 904 Based on your previous
 905 answers, provide an
 906 answer to the last question
 907 in a JSON format with
 908 only keys: "reasoning"
 909 and "agreement".
 910
 911 Interviewer: What is your
 912 name?
 913 Interviewee: I am
 914 {name_prefix} {name}.
 915 Interviewer: Where do you
 916 live?
 917 Interviewee: I live in the
 918 U.S.
 919 Interviewer: How old are you?
 920 Interviewee: I'm {age} years
 921 old.
 922 Interviewer: What is your
 923 gender?
 924 Interviewee: I'm {sex}.
 925 Interviewer: How would you
 926 describe your ethnicity?
 927 Interviewee: I'm {ethnicity}.
 928 Interviewer: What's your
 929 annual income?

930 Interviewee: {income} per
 931 year.
 932 Interviewer: Do you have a
 933 university degree?
 934 Interviewee: {education}
 935 Interviewer: Do you identify
 936 with a political party
 937 in the U.S.?
 938 Interviewee: {polview}
 939 Interviewer: Tell me if you
 940 disagree or agree with
 941 the following
 942 statement:
 943 "{claim_text}". First
 944 explain your reasoning
 945 in three sentences and
 946 then give your answer
 947 by choosing one of
 948 ["strongly disagree",
 949 "disagree", "agree",
 950 "strongly agree"].
 951 Interviewee: ..
 952 {post}''

953 D Model details

954 E Detailed Results

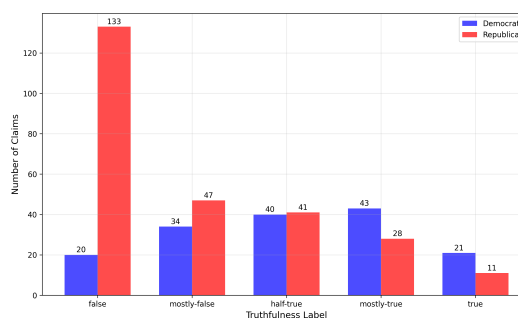


Figure 8: Distributions in Politifact

Model	Source	Size [B]	License
Mistral 7B	mistralai/Mistral-7B-Instruct-v0.1 ^a	7	Apache-2.0
Mixtral 8x7B	mistralai/Mixtral-8x7B-Instruct-v0.1 ^a	56	Apache-2.0
Qwen3 4B	Qwen/Qwen3-4B-Instruct-2507 ^a	4	Apache-2.0
Qwen3 30B	Qwen/Qwen3-30B-A3B-Instruct-2507 ^a	72	Apache-2.0
Llama 3 8B	meta-llama/Meta-Llama-3-8B-Instruct ^a	8	llama3
Llama 4 Scout	meta-llama/Llama-4-Scout-17B-16E-Instruct ^a	109	llama4
GPT-OSS 20B	openai/gpt-oss-20b ^a	20	Apache-2.0
GPT 4.1	openai/gpt-4.1 ^b	200 ^c	paid service
DeepSeek R1 Dist.	deepseek-ai/DeepSeek-R1-Distill-Qwen-32B ^a	32	MIT
Deepseek 3.1	deepseek/deepseek-chat-v3.1:free ^b	671	paid service
Grok 3	x-ai/grok-3 ^b	500 ^d	paid service
Grok 4	x-ai/grok-4-fast:free ^b	1700 ^d	paid service
Phi 3 mini 4K	unsloth/Phi-3-mini-4k-instruct ^a	3.8	access upon request
Phi Conservative	Opinion-GPT/opiniongpt-phi3-conservative ^a	3.8	access upon request
Phi Liberal	Opinion-GPT/opiniongpt-phi3-liberal ^a	3.8	access upon request

^a From <https://huggingface.co/>.

^b From <https://openrouter.ai/>.

^c Estimated size from <https://aiexpjourney.substack.com/p/the-number-of-parameters-of-gpt-4o>.

^d Estimated size from <https://www.datastudios.org/post/grok-4-vs-previous-models>.

Table 5: Overview of LLMs, their sources, parameter sizes, and licenses.

LLM	Corr(PB, TB)	Corr(PB, F1)
Phi 3 mini 4K	0.472	0.251
Phi 3 Liberal	0.642	-0.138
Phi 3 Conservative	0.728	-0.019
Mistral 7B	0.86	-0.083
Mixtral 8x7B	0.779	0.045
Qwen3 4B	0.891	-0.125
Qwen3 30B	0.864	0.126
Llama3 8B	0.28	0.219
Llama 4 Scout	0.952	0.392
GPT-OSS 20B	0.038	-0.184
GPT 4.1	0.531	0.279
DeepSeek R1 Dist.	0.889	-0.185
DeepSeek 3.1	0.743	-0.437
Grok 3	0.128	-0.025
Grok 4 fast	0.281	-0.106
Average	0.605	0.001

Table 6: Correlations between Political Bias and Performance Metrics (LLM with personas)

LLM	Prompt	Axis	P-values					
			Sex	Ethnicity	Education	Political View	Age	Income
Phi 3 mini 4K	1	economic	0.290	0.001	0.068	0.119	0.833	0.253
Phi 3 mini 4K	1	social	0.623	0.000	0.119	0.000	0.955	0.335
Phi 3 mini 4K	1	both	0.317	0.000	0.021	0.000	0.951	0.874
Phi 3 mini 4K	2	economic	0.521	0.429	0.418	0.000	0.100	0.276
Phi 3 mini 4K	2	social	0.843	0.035	0.038	0.004	0.501	0.108
Phi 3 mini 4K	2	both	0.854	0.062	0.033	0.000	0.752	0.349
Phi 3 Liberal	1	economic	0.781	0.049	0.000	0.000	0.728	0.244
Phi 3 Liberal	1	social	0.652	0.064	0.000	0.247	0.353	0.038
Phi 3 Liberal	1	both	0.875	0.024	0.000	0.139	0.524	0.416
Phi 3 Liberal	2	economic	0.897	0.363	0.530	0.000	0.359	0.000
Phi 3 Liberal	2	social	0.622	0.098	0.157	0.000	0.264	0.029
Phi 3 Liberal	2	both	0.726	0.554	0.473	0.000	0.765	0.036
Phi 3 Conservative	1	economic	0.802	0.194	0.127	0.006	0.622	0.006
Phi 3 Conservative	1	social	0.899	0.042	0.512	0.000	0.392	0.570
Phi 3 Conservative	1	both	0.951	0.020	0.117	0.000	0.694	0.169
Phi 3 Conservative	2	economic	0.124	0.439	0.183	0.000	0.669	0.144
Phi 3 Conservative	2	social	0.015	0.085	0.345	0.000	0.162	0.992
Phi 3 Conservative	2	both	0.007	0.056	0.487	0.000	0.668	0.377
Mistral 7B	1	economic	0.381	0.566	0.645	0.709	0.277	0.961
Mistral 7B	1	social	0.810	0.486	0.468	0.040	0.037	0.131
Mistral 7B	1	both	0.771	0.218	0.313	0.222	0.406	0.186
Mistral 7B	2	economic	0.343	0.963	0.722	0.000	0.882	0.170
Mistral 7B	2	social	0.800	0.008	0.477	0.000	0.019	0.350
Mistral 7B	2	both	0.621	0.083	0.691	0.000	0.105	0.742
Mixtral 8x7B	1	economic	0.195	0.119	0.648	0.056	0.094	0.002
Mixtral 8x7B	1	social	0.301	0.030	0.041	0.000	0.001	0.518
Mixtral 8x7B	1	both	0.120	0.081	0.053	0.000	0.001	0.017
Mixtral 8x7B	2	economic	0.074	0.230	0.403	0.000	0.381	0.697
Mixtral 8x7B	2	social	0.836	0.358	0.514	0.000	0.320	0.553
Mixtral 8x7B	2	both	0.329	0.435	0.662	0.000	0.105	0.425
Qwen3 4B	1	economic	0.907	0.075	0.318	0.064	0.365	0.695
Qwen3 4B	1	social	0.102	0.201	0.914	0.574	0.494	0.437
Qwen3 4B	1	both	0.180	0.626	0.564	0.073	0.168	0.362
Qwen3 4B	2	economic	0.913	0.160	0.632	0.000	0.867	0.015
Qwen3 4B	2	social	0.845	0.004	0.671	0.000	0.336	0.036
Qwen3 4B	2	both	0.826	0.005	0.570	0.000	0.770	0.001
Qwen3 30B	1	economic	0.098	0.056	0.852	0.000	0.626	0.289
Qwen3 30B	1	social	0.000	0.000	0.143	0.001	0.004	0.840
Qwen3 30B	1	both	0.000	0.000	0.231	0.000	0.132	0.592
Qwen3 30B	2	economic	0.309	0.471	0.313	0.000	0.303	0.069
Qwen3 30B	2	social	0.023	0.173	0.187	0.000	0.473	0.383
Qwen3 30B	2	both	0.034	0.343	0.872	0.000	0.309	0.095
Llama3 8B	1	economic	0.185	0.856	0.422	0.000	0.067	0.533
Llama3 8B	1	social	0.696	0.703	0.056	0.003	0.390	0.039
Llama3 8B	1	both	0.281	0.661	0.482	0.000	0.178	0.057
Llama3 8B	2	economic	0.264	0.921	0.051	0.000	0.117	0.260
Llama3 8B	2	social	0.081	0.203	0.240	0.000	0.014	0.712

Continued on next page

LLM	Prompt	Axis	P-values					
			Sex	Ethnicity	Education	Political View	Age	Income
Llama3 8B	2	both	0.351	0.310	0.316	0.000	0.010	0.428
Llama 4 Scout 17B	1	economic	0.000	0.074	0.223	0.000	0.611	0.105
Llama 4 Scout 17B	1	social	0.000	0.894	0.234	0.000	0.044	0.000
Llama 4 Scout 17B	1	both	0.000	0.241	0.104	0.000	0.366	0.000
Llama 4 Scout 17B	2	economic	0.911	0.017	0.009	0.000	0.710	0.042
Llama 4 Scout 17B	2	social	0.001	0.887	0.256	0.000	0.817	0.003
Llama 4 Scout 17B	2	both	0.015	0.375	0.125	0.000	0.834	0.001
GPT-OSS 20B	1	economic	0.302	0.004	0.076	0.574	0.296	0.580
GPT-OSS 20B	1	social	0.425	0.351	0.618	0.030	0.198	0.649
GPT-OSS 20B	1	both	0.232	0.031	0.381	0.036	0.237	0.508
GPT-OSS 20B	2	economic	0.303	0.130	0.177	0.008	0.265	0.508
GPT-OSS 20B	2	social	0.158	0.860	0.143	0.001	0.311	0.637
GPT-OSS 20B	2	both	0.088	0.269	0.130	0.509	0.973	0.439
GPT 4.1	1	economic	0.305	0.944	0.278	0.000	0.440	0.603
GPT 4.1	1	social	0.004	0.700	0.544	0.000	0.084	0.492
GPT 4.1	1	both	0.012	0.867	0.693	0.000	0.211	0.969
GPT 4.1	2	economic	0.284	0.575	0.556	0.000	0.445	0.734
GPT 4.1	2	social	0.852	0.096	0.064	0.000	0.027	0.057
GPT 4.1	2	both	0.603	0.356	0.453	0.000	0.072	0.090
DeepSeek R1 Distill	1	economic	0.286	0.807	0.689	0.000	0.445	0.312
DeepSeek R1 Distill	1	social	0.286	0.246	0.336	0.000	0.430	0.754
DeepSeek R1 Distill	1	both	0.812	0.593	0.510	0.000	0.232	0.416
DeepSeek R1 Distill	2	economic	0.815	0.982	0.206	0.000	0.839	0.766
DeepSeek R1 Distill	2	social	0.223	0.314	0.240	0.000	0.386	0.196
DeepSeek R1 Distill	2	both	0.460	0.563	0.093	0.000	0.702	0.262
DeepSeek 3.1	1	economic	0.750	0.039	0.005	0.000	0.296	0.151
DeepSeek 3.1	1	social	0.004	0.001	0.015	0.000	0.946	0.367
DeepSeek 3.1	1	both	0.076	0.053	0.002	0.000	0.577	0.163
DeepSeek 3.1	2	economic	0.798	0.731	0.036	0.000	0.879	0.467
DeepSeek 3.1	2	social	0.248	0.130	0.074	0.000	0.209	0.773
DeepSeek 3.1	2	both	0.483	0.410	0.033	0.000	0.405	0.924
Grok 3	1	economic	0.004	0.024	0.002	0.000	0.906	0.091
Grok 3	1	social	0.001	0.536	0.186	0.000	0.219	0.011
Grok 3	1	both	0.613	0.270	0.001	0.000	0.478	0.431
Grok 3	2	economic	0.709	0.380	0.044	0.000	0.501	0.004
Grok 3	2	social	0.033	0.373	0.418	0.000	0.115	0.219
Grok 3	2	both	0.059	0.280	0.071	0.000	0.498	0.466
Grok 4 fast	1	economic	0.764	0.699	0.781	0.752	0.774	0.939
Grok 4 fast	1	social	0.397	0.958	0.275	0.065	0.732	0.969
Grok 4 fast	1	both	0.413	0.943	0.551	0.118	0.818	0.980
Grok 4 fast	2	economic	0.888	0.068	0.343	0.178	0.353	0.795
Grok 4 fast	2	social	0.064	0.806	0.711	0.000	0.486	0.398
Grok 4 fast	2	both	0.251	0.094	0.883	0.000	0.206	0.704

LLM	Prompt	Axis	Mean Error (ME)			Mean Absolute Error (MAE)		
			Rep	Dem	P-value	Rep	Dem	P-value
Phi 3 Conservative	1	both	-0.8011	-0.4960	0.1039	1.1924	1.1555	0.8478
Phi 3 Conservative	1	economic	-0.6797	-0.4307	0.3194	1.0911	1.0580	0.8789
Phi 3 Conservative	1	social	-0.8683	-0.5322	0.1930	1.2485	1.2096	0.8148
Phi 3 Conservative	2	both	-0.7751	-0.4126	0.0442	1.1557	1.0821	0.9514
Phi 3 Conservative	2	economic	-0.6743	-0.2273	0.1111	1.0583	1.0701	0.6618
Phi 3 Conservative	2	social	-0.8309	-0.5153	0.1924	1.2096	1.0888	0.8061
DeepSeek 3.1	1	both	-0.3953	0.0275	0.0003	0.8442	0.7547	0.4129
DeepSeek 3.1	1	economic	-0.4837	-0.0196	0.0131	0.8815	0.7370	0.4031
DeepSeek 3.1	1	social	-0.3464	0.0536	0.0092	0.8235	0.7645	0.7650
DeepSeek 3.1	2	both	-0.3961	0.0140	0.0004	0.8101	0.7729	0.5749
DeepSeek 3.1	2	economic	-0.4902	0.0217	0.0111	0.8098	0.7804	0.7935
DeepSeek 3.1	2	social	-0.3440	0.0096	0.0111	0.8102	0.7687	0.6154
DeepSeek R1 Dist.	1	both	-0.5480	-0.2246	0.0217	0.9679	0.9201	0.6717
DeepSeek R1 Dist.	1	economic	-0.5860	-0.4100	0.2494	0.9657	0.8765	0.4895
DeepSeek R1 Dist.	1	social	-0.5269	-0.1219	0.0458	0.9691	0.9442	0.9318
DeepSeek R1 Dist.	2	both	-0.6245	-0.1214	0.0011	0.9466	0.9354	0.7167
DeepSeek R1 Dist.	2	economic	-0.6725	-0.2798	0.0609	0.9741	0.9011	0.6876
DeepSeek R1 Dist.	2	social	-0.5979	-0.0337	0.0084	0.9314	0.9543	0.4373
GPT-OSS 20B	1	both	-1.3191	-0.9291	0.0117	1.4875	1.2428	0.0849
GPT-OSS 20B	1	economic	-1.4047	-1.1091	0.1645	1.6349	1.3423	0.1183
GPT-OSS 20B	1	social	-1.2717	-0.8293	0.0320	1.4058	1.1876	0.3121
GPT-OSS 20B	2	both	-1.2518	-0.8548	0.0137	1.4004	1.1642	0.1345
GPT-OSS 20B	2	economic	-1.3875	-0.9787	0.0947	1.4627	1.1974	0.1792
GPT-OSS 20B	2	social	-1.1765	-0.7861	0.0829	1.3659	1.1458	0.4307
GPT 4.1	1	both	-0.4705	-0.2891	0.1873	0.7690	0.8915	0.2721
GPT 4.1	1	economic	-0.5087	-0.1946	0.1891	0.8565	0.9663	0.3500
GPT 4.1	1	social	-0.4494	-0.3416	0.4377	0.7205	0.8500	0.6075
GPT 4.1	2	both	-0.4640	-0.2795	0.1765	0.7686	0.9058	0.2391
GPT 4.1	2	economic	-0.5087	-0.1848	0.1472	0.8565	1.0217	0.1675
GPT 4.1	2	social	-0.4392	-0.3319	0.4942	0.7199	0.8416	0.7434
Grok 4 fast	1	both	-0.2616	0.2372	0.0006	0.7547	0.9225	0.1352
Grok 4 fast	1	economic	-0.2772	0.3750	0.0037	0.7533	0.9250	0.1777
Grok 4 fast	1	social	-0.2530	0.1608	0.0382	0.7554	0.9211	0.3812
Grok 4 fast	2	both	-0.2434	0.2301	0.0010	0.7977	0.8706	0.6856
Grok 4 fast	2	economic	-0.2935	0.3326	0.0070	0.8370	0.8870	0.6903
Grok 4 fast	2	social	-0.2157	0.1733	0.0464	0.7759	0.8616	0.8412
Grok 3	1	both	-0.3860	-0.1426	0.0853	0.7333	0.8372	0.2658
Grok 3	1	economic	-0.4826	-0.2000	0.1967	0.7957	0.8783	0.4304
Grok 3	1	social	-0.3325	-0.1108	0.2972	0.6988	0.8145	0.4320
Grok 3	2	both	-0.4109	-0.1318	0.0383	0.7147	0.8171	0.2636
Grok 3	2	economic	-0.4739	-0.0696	0.0770	0.7261	0.9043	0.1604
Grok 3	2	social	-0.3759	-0.1663	0.2264	0.7084	0.7687	0.7409
Phi 3 Liberal	1	both	-0.9621	-0.7123	0.2939	1.2368	1.2401	0.6802
Phi 3 Liberal	1	economic	-1.0449	-0.6187	0.1730	1.4015	1.3335	0.6478
Phi 3 Liberal	1	social	-0.9162	-0.7642	0.7688	1.1455	1.1883	0.4324
Phi 3 Liberal	2	both	-0.8438	-0.4738	0.0299	1.1586	1.1205	0.9024
Phi 3 Liberal	2	economic	-0.8769	-0.2285	0.0113	1.1972	1.0751	0.3712

Continued on next page

LLM	Prompt	Axis	Mean Error (ME)			Mean Absolute Error (MAE)		
			Rep	Dem	P-value	Rep	Dem	P-value
Phi 3 Liberal	2	social	-0.8255	-0.6098	0.4372	1.1371	1.1457	0.5350
Llama3 8B	1	both	-0.7465	-0.4508	0.0543	1.1132	0.9647	0.1733
Llama3 8B	1	economic	-0.8772	-0.5837	0.1610	1.1853	0.9453	0.0902
Llama3 8B	1	social	-0.6741	-0.3771	0.1936	1.0733	0.9755	0.6995
Llama3 8B	2	both	-0.9303	-0.6194	0.0447	1.1978	1.0100	0.1293
Llama3 8B	2	economic	-1.0443	-0.7246	0.0887	1.2134	0.9848	0.1240
Llama3 8B	2	social	-0.8671	-0.5612	0.1953	1.1892	1.0240	0.4440
Llama 4 Scout	1	both	-0.1687	0.1048	0.0460	0.7528	0.8678	0.2369
Llama 4 Scout	1	economic	-0.1287	-0.0073	0.6466	0.6632	0.8274	0.3373
Llama 4 Scout	1	social	-0.1900	0.1649	0.0379	0.8004	0.8895	0.3763
Llama 4 Scout	2	both	-0.1912	0.2397	0.0043	0.7716	0.9901	0.0156
Llama 4 Scout	2	economic	-0.2374	0.1230	0.1002	0.6731	1.0104	0.0219
Llama 4 Scout	2	social	-0.1656	0.3044	0.0174	0.8262	0.9789	0.1812
Mistral 7B	1	both	-0.6322	-0.6480	0.9707	1.0638	1.0286	0.5669
Mistral 7B	1	economic	-0.7721	-0.7727	0.9813	1.0069	1.0206	0.7693
Mistral 7B	1	social	-0.5547	-0.5789	0.9678	1.0953	1.0331	0.6204
Mistral 7B	2	both	-0.7669	-0.6672	0.4058	1.1253	1.0101	0.2626
Mistral 7B	2	economic	-0.8315	-0.6692	0.2365	1.0243	0.9561	0.4602
Mistral 7B	2	social	-0.7310	-0.6661	0.8061	1.1812	1.0400	0.3963
Mixtral 8x7B	1	both	-0.8199	-0.5250	0.0638	1.0632	1.0175	0.4131
Mixtral 8x7B	1	economic	-0.9660	-0.6358	0.1303	1.1514	1.0475	0.2853
Mixtral 8x7B	1	social	-0.7397	-0.4613	0.2295	1.0148	1.0002	0.9399
Mixtral 8x7B	2	both	-0.7987	-0.2593	0.0018	1.0563	0.9398	0.2731
Mixtral 8x7B	2	economic	-0.9559	-0.3367	0.0089	1.0871	0.8867	0.1948
Mixtral 8x7B	2	social	-0.7131	-0.2156	0.0492	1.0395	0.9698	0.6373
Phi 3 mini 4K	1	both	-0.4621	-0.1189	0.1312	0.9924	1.0252	0.3211
Phi 3 mini 4K	1	economic	-0.3681	-0.0513	0.2246	0.8204	0.8343	0.8087
Phi 3 mini 4K	1	social	-0.5142	-0.1564	0.2672	1.0878	1.1310	0.3439
Phi 3 mini 4K	2	both	-0.6127	-0.2240	0.0471	1.0354	0.9966	0.5863
Phi 3 mini 4K	2	economic	-0.5022	0.0152	0.0456	1.0319	0.8870	0.6282
Phi 3 mini 4K	2	social	-0.6740	-0.3566	0.2698	1.0374	1.0574	0.3462
Qwen3 4B	1	both	-1.3897	-1.1685	0.1544	1.4434	1.3176	0.4651
Qwen3 4B	1	economic	-1.4359	-1.1757	0.3209	1.5337	1.3641	0.4437
Qwen3 4B	1	social	-1.3641	-1.1645	0.2716	1.3934	1.2918	0.6927
Qwen3 4B	2	both	-1.3430	-0.9981	0.0329	1.3957	1.1797	0.1941
Qwen3 4B	2	economic	-1.3692	-0.9743	0.1529	1.4562	1.1938	0.2953
Qwen3 4B	2	social	-1.3285	-1.0112	0.1067	1.3622	1.1719	0.3691
Qwen3 30B	1	both	-1.0066	-0.6644	0.0352	1.1658	1.0918	0.7543
Qwen3 30B	1	economic	-0.8528	-0.6673	0.4463	1.0866	1.0967	0.9408
Qwen3 30B	1	social	-1.0918	-0.6629	0.0478	1.2096	1.0890	0.7078
Qwen3 30B	2	both	-1.0135	-0.5440	0.0021	1.2182	1.0401	0.2529
Qwen3 30B	2	economic	-0.9048	-0.5285	0.1043	1.1857	1.0385	0.4993
Qwen3 30B	2	social	-1.0738	-0.5526	0.0103	1.2363	1.0410	0.3751

LLM	No Persona						Democrat Persona						Republican Persona						Persona with No Political View					
	PB	TB ₁	F1 ₁	TB ₂	F1 ₂		PB	TB ₁	F1 ₁	TB ₂	F1 ₂		PB	TB ₁	F1 ₁	TB ₂	F1 ₂		PB	TB ₁	F1 ₁	TB ₂	F1 ₂	
Phi 3 Conservative	-0.397	-0.305	0.225	-0.362	0.218		-0.446	-0.138	0.156	-0.499	0.190		-0.387	-0.102	0.154	-0.106	0.186		-0.449	-0.067	0.148	-0.274	0.187	
DeepSeek 3.1	-0.210	-0.423	0.368	-0.410	0.358		-0.029	-0.528	0.322	-0.772	0.310		0.233	0.025	0.282	0.607	0.280		-0.041	-0.316	0.314	-0.439	0.314	
DeepSeek R1 Dist.	-0.413	-0.323	0.245	-0.503	0.285		-0.475	-0.554	0.281	-0.689	0.280		-0.197	-0.470	0.292	-0.439	0.272		-0.395	-0.466	0.280	-0.560	0.270	
GPT-OSS 20B	-0.341	-0.390	0.160	-0.397	0.173		-0.335	-0.236	0.128	-0.237	0.124		-0.247	-0.262	0.121	-0.238	0.116		-0.310	-0.238	0.129	-0.246	0.131	
GPT 4.1	0.054	-0.181	0.425	-0.184	0.419		0.127	-0.314	0.414	-0.518	0.429		0.389	-0.183	0.424	-0.404	0.439		0.121	-0.209	0.397	-0.463	0.419	
Grok 4	-0.091	-0.499	0.403	-0.474	0.398		-0.276	-0.524	0.305	-0.595	0.304		-0.043	-0.484	0.291	-0.475	0.299		-0.265	-0.453	0.296	-0.475	0.305	
Grok 3	0.187	-0.243	0.433	-0.279	0.430		0.093	-0.449	0.383	-0.482	0.364		0.092	-0.271	0.361	-0.268	0.355		0.081	-0.330	0.376	-0.334	0.339	
Phi 3 Liberal	-0.397	-0.250	0.209	-0.370	0.216		-0.430	-0.474	0.212	-1.036	0.226		-0.390	-0.424	0.206	-0.486	0.226		-0.433	-0.437	0.208	-0.819	0.233	
Llama3 8B	-0.393	-0.296	0.160	-0.311	0.142		-0.150	-0.482	0.248	-0.914	0.237		-0.092	-0.448	0.243	-0.645	0.242		-0.184	-0.411	0.231	-0.634	0.227	
Llama 4 Scout	-0.451	-0.274	0.346	-0.431	0.332		-0.563	-0.347	0.324	-0.724	0.282		-0.101	-0.184	0.308	-0.208	0.300		-0.452	-0.198	0.312	-0.447	0.299	
Mistral 7B	-0.251	0.016	0.195	-0.100	0.191		-0.210	-0.052	0.198	-0.432	0.208		-0.119	-0.063	0.201	-0.158	0.207		-0.175	-0.056	0.212	-0.272	0.204	
Mistral 8x7B	-0.377	-0.295	0.279	-0.539	0.288		-0.284	-0.420	0.254	-0.643	0.242		-0.131	-0.406	0.267	-0.379	0.244		-0.274	-0.378	0.249	-0.496	0.244	
Phi 3 mini 4K	-0.467	-0.343	0.320	-0.389	0.273		-0.414	-0.310	0.255	-0.556	0.248		-0.375	-0.404	0.259	-0.434	0.259		-0.402	-0.365	0.259	-0.465	0.248	
Qwen3 4B	-0.434	-0.221	0.144	-0.345	0.167		-0.516	-0.358	0.224	-0.496	0.218		-0.217	-0.344	0.224	-0.358	0.213		-0.489	-0.365	0.232	-0.495	0.214	
Qwen3 30B	-0.494	-0.342	0.268	-0.470	0.272		-0.608	-0.515	0.298	-0.667	0.269		-0.294	-0.475	0.289	-0.453	0.273		-0.586	-0.467	0.294	-0.596	0.272	

Table 9: Overview of LLM Performance by Persona Type